impl TradeoffCurve::from_gdp

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This proof resides in **"contrib"** because it has not completed the vetting process.

1 Hoare Triple

Precondition

Compiler-verified

- Argument `mu` of type `f64`

Caller-verified

None.

Pseudocode

```
def tradeoffcurve_from_gdp(mu: float) -> TradeoffCurve:
    sqrt_2_lo = next_down(sqrt(2.0))
    p_max = next_down_f32(2.0)
    t_max = f32_max

    def tradeoff(alpha: float) -> float:
        if alpha == 0.0:
            return 1.0
        if alpha == 1.0:
            return 0.0

        p = neg_inf_mul(neg_inf_sub(1.0, alpha), 2.0)
        p = min(neg_inf_cast_f32(p), p_max)
        if p == 0.0:
            return 0.0

        y = next_up_f32(inf_cast_f32(erfc_inv(p)))
        q = inf_div(mu, sqrt_2_lo)
        t = inf_add(y, q)
        if t > t_max:
            return 0.0

        t = inf_cast_f32(t)
        beta_num = max(next_down_f32(neg_inf_cast_f32(erfc(t))), 0.0)
        return neg_inf_div(beta_num, 2.0)

    return TradeoffCurve(tradeoff)
```

Postcondition

Theorem 1.1. Given $\mu \geq 0$, the pseudocode returns a conservative implementation of the Gaussian differential privacy tradeoff curve

$$\beta(\alpha) = \Phi\big(\Phi^{-1}(1-\alpha) - \mu\big)\,.$$

Proof. Using $\Phi(x) = \frac{1}{2}\operatorname{erfc}(-x/\sqrt{2})$, the GDP tradeoff curve can be written as

$$\beta(\alpha) = \tfrac{1}{2}\,\operatorname{erfc}\Big(\operatorname{erfc}^{-1}(2(1-\alpha)) + \mu/\sqrt{2}\Big)\,,$$

as defined in Theorem 2.5 of Dong, Roth, and Su 2022.

The pseudocode rounds the input to erfc^{-1} downward, then rounds the output upward; because erfc^{-1} is decreasing, this can only increase the final argument to erfc. It also rounds $\mu/\sqrt{2}$ upward and rounds the final erfc value downward. Since erfc is decreasing, each of these choices can only decrease the returned $\beta(\alpha)$. Therefore the implementation is conservative. The endpoint returns at $\alpha = 0$ and $\alpha = 1$ match the exact GDP tradeoff curve. □

References

Dong, Jinshuo, Aaron Roth, and Weijie J. Su (2022). "Gaussian Differential Privacy". In: *Journal of the Royal Statistical Society Series B: Statistical Methodology* 84.1, pp. 3–37. ISSN: 1369-7412.